

Using Eye Gaze for Navigation: A User Study Inspired by Lucid Dreams

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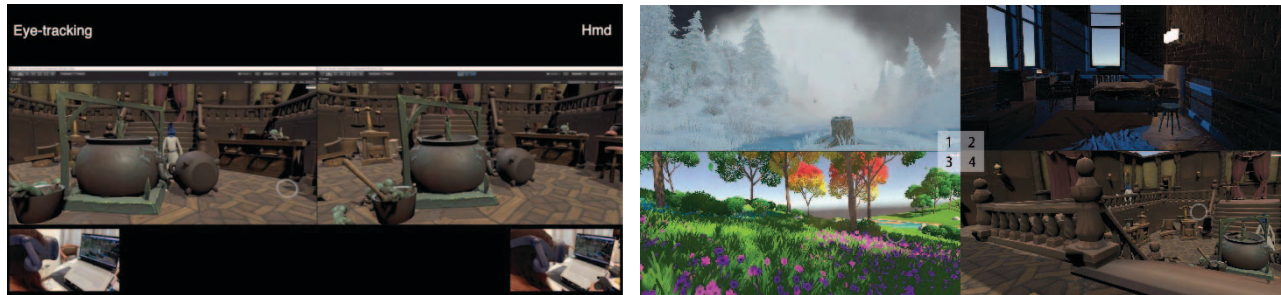


Figure 1. Left: Screenshot of two different navigations methods. Right: Scenario design of four scenes.

ABSTRACT

We implemented a new navigation method using Eye-tracking to steer the direction and compared our method with using HMD Head-ray for steering. Our idea was inspired by rapid eyeball movement (REM) often observed in lucid dreams. A within-group experiment was conducted where participants experienced four different VR scenarios with both navigation methods. The results suggested that using Eye-gaze as a navigation method is viable in VR (similar travel distance, speed, and time), and that the angle between the Eye-gaze ray and the Head-ray is significantly higher in all four scenes with the Eye-tracking steering method than the HMD Head-ray one.

Keywords: Virtual reality, lucid dreams, eye-tracking.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, Augmented, and Virtual Realities; J.4 [Computer Applications]: Social and Behavioral Sciences—Psychology

1. INTRODUCTION

The study of dreams could provide researchers with clues to create more immersive virtual environments. Indeed, a closer study of how we “navigate” in our dreams might provide a template for designing more engaging, and perhaps less nauseous-susceptible way of navigation in the virtual world. Interestingly, most of our dreams occur during Rapid Eye Movement (REM) sleep, where our eyeballs move rapidly while our limbs stay more or less motionless [1]. According to LaBerge's theory [2], this time is also a period of highly active human consciousness. In this work, we are interested recreating the physiological experienced in dreams, using VR.

Specifically, we used eye-tracking technology to allow users to navigate in VR, thereby simulating the feeling of dreaming.

2. BACKGROUND

Lucid dreaming is an elusive dream state where dreamers are able to act consciously upon reflection, and even to control the dream to some degree [3]. This fascinating state has been an ongoing topic of exploration in the neuroscience community, with more and more researchers focusing on combining virtual reality technology with lucid dreaming for their research. Gott et al. [3] found that, with virtual reality training, some were able to have lucid dreams more often than untrained dreamers, to the same effect as the traditional training.

According to LaBerge [4], lucid dreaming usually (though perhaps not exclusively) occurs during REM sleep. This concept guided our design philosophy to using Eye-tracking technology for creating a new method for steering navigation in the virtual environment. Previous research in VR locomotion demonstrated the advantage of using the eye-tracking method to teleport, resulting in less distraction for users [5]. Spurgeon [6] explored blinking for teleportation, the results revealed that this method performed well in terms of performance, accuracy, and learnability. These previous studies shown that using Eye-tracking for teleportation in virtual environment is viable. However, in this work we are interested in simulation dream-like experiences where users could move continuously in VR, rather jumping from one point to another like in teleportation.

3. EXPERIMENTAL DESIGN

This is a within-group study. We designed four scenes, and invited participants to experience all four scenes (all in the same order, S1 to S4, see Section 3.3) twice, each time with one of the two navigation methods (Eye-Gaze Steering & Head-ray Steering).

3.1 Hardware & Software Specification

We developed our VR project with *HTC VIVE PRO EYE* headset in *Unity3D* game engine with the programming language *C#*. We used *SteamVR sdk v1.13.10*, *VR Toolkit v3.3.0*, *Tobii XR sdk v1.0.0* and *SRanipal v1.0.1* for creating the interaction features.

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3.2 The Navigation Methods

Eye-Gaze Steering: Players would move along the direction of combined gaze ray where the eyes converge, when they press the VR Controller's right button of Touchpad. The Gaze Visualizer enable a circle to encapsulate people's gaze which is always on display in this condition (See Figure 1. Left). **Head-ray Steering:** When players press the left button of Touchpad, they would move along the direction of the forward ray emitted by the headset. Participants can switch the mode of steering by pressing left or right button of VR Controller.

3.3 Scenarios

Four scenes (see Figure 1, right) were designed and implemented based on a recurrent dream experience by one of our team members. In her dreams, she constantly moved between several strange yet fascinating scenarios. During the process of completing tasks, participants also interacted with some non-player characters (NPCs) which were designed in order to fully restore the experience of the dream, and also to assist us in testing the feasibility of our new navigation method.

Scene1(S1) Participants would move through a blizzard environment where they need to find a house, with the wind hindering their progress.

Scene2(S2) Participants would be in a bedroom with little space and needed to find clues to entry to the next scene.

Scene3(S3) Participants would arrive in the middle of a picturesque forest and their goal was to find a red house.

Scene4(S4) Participants would find themselves inside a topographically complex interior with two floors and two rooms. They would be asked to rescue a child, and at the same time, the witches in the room would start chasing the players. Their goal would be to rescue the child and escape from the room.

3.4 Participants and Procedure

Ten participants (6 females, age 36.3 ± 10.5) attended our study. Six of them never experienced VR, and 4 had some prior experience. Participants were asked to experience the scenarios (S1-S4, in that order) twice, each time with a different navigation method. Five participants tested the Eye-Gaze first, then the Head-ray. The other five participants completed this process in the reverse order. The purpose of this experiment is to measure which navigation method will trigger the eyeballs to become more active. We recorded the absolute value of the *Angle* between the ray emitted from the headset and the eye-tracking ray per second via Unity 3D. We also recorded participants' *Distance* and *Time* of travel for each scene, with each method.

4. RESULTS

In order to compare the two navigation methods, we ran paired t-tests for *Distance*, *Time*, *Speed* (calculated with distance travelled and time spent), and *Angle* in each scene. No significance was found for *Distance*, *Time*, and *Speed* other than a small effect of *Time* in Scene 4 ($p = 0.031$, as shown in Figure 2, participants tended to spend longer time with the Eye-gaze method in the last scene). This indicated that overall, there is no major differences in these two ways of steering in navigation. However, it was very interesting to observe a very consistent effect for *Angle* across all scenes (S1, S2, S4: $p = 0.000$, S3: $p = 0.002$). As shown in figure 2, it is clear that the absolute angle between eye-ray and head-ray is bigger in Eye-tracking steering method than HMD Head-ray steering method. This result indicates that participants moved their eyes a lot more (average 23 degrees for the Eye-tracking method, 14 degrees for the Head-ray method) in the Eye-tracking method.

5. CONCLUSIONS AND FUTURE WORK

In this study we compared the behavioural data recorded from participants who went through a dream-like experience with two different navigation methods, namely Eye-gaze steering and Head-ray steering. We designed virtual environments with both even and uneven planes, and a variety of tasks for players to explore, including interactions with a non-player character. Further, we used a single device to evaluate the user experience. Our result provided evidence supporting the feasibility of using eye tracking for steering navigation in virtual environment, as overall there was no major difference in terms of participants' travel distance, time, and speed. Secondly, we found that with the Eye-tracking steering method, participants' eyeball activity is significantly higher than using HMD for steering across all four scenes. Our method could contribute towards future research in simulating REM cycles observed during dreaming periods. In future, similar navigation methods using eye-tracking could be tested with participants lying in bed while wearing an HMD. We hope our work could inspire novel and more efficient VR trainings for inducing lucid dreaming state where we could control our dream content, in comparison to existing traditional training. Further, we have observed that when turning back, most participants opted to use a combination of head and eye rotation. In order to elicit lucid dreaming state, future technical work could look at improving the implementation to offer purely gaze-controlled navigation methods. This is because during dreaming people do not usually physically rotate their head.

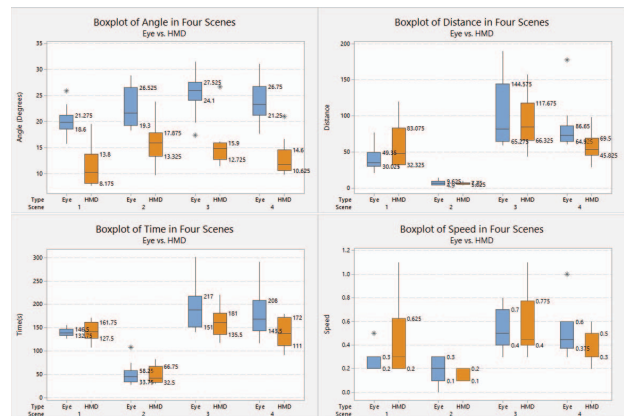


Figure 2. Boxplot of Angle, Distance, Time, and Speed. Blue: Eye-tracking for steering; Orange: HMD Head-ray for steering.

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